VISUAL ANALYTICS

RENTAL PRICES IN USA REPORT

BY

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INTRODUCTION

- In this project, we analyzed rental price trends across the United States using a large dataset. Rental costs play a big role in financial decisions for both people and businesses, so it's important to understand the trends and reasons behind changes in rent.
- We used **Tableau** to explore the data, look at current rental prices, and try to predict future trends. This report also identifies the main factors that affect rent changes.
- Join us as we explore rental patterns across the U.S. and provide insights that could help renters, property owners, and real estate professionals make better decisions.

EXPLORE THE DATA

We used Tableau to effectively evaluate and analyze the rental data by utilizing a variety of strategies and producing a number of visually appealing graphics. Below is a list of the charts I made, along with an explanation of each:

1. Bar Chart: State-by-State Average Rent The average rental prices in the several US states are contrasted in this bar graph. It helps determine which states have the highest and lowest rental costs.
Rental Trends Over Time: A bar graph that illustrates how rental costs have changed over time. This graphic is crucial for identifying trends and seasonality in rental pricing.

2. Line Chart: Differences in rent costs by city are shown in line charts. The historical variations in rental prices in a few selected cities are depicted in this line graph. It makes it simpler to see how rental costs vary depending on the area.

Yearly Rent Growth: The rental price growth rate on an annual basis is displayed in a line graph. Finding periods of noticeable expansion or contraction in the rental market requires the use of this graph.

3. Heat Map: This heat map displays rental prices in various ZIP codes, giving a geographic perspective on rental expenses.

Seasonal Rent Variations: This heat map illustrates how rental costs fluctuate by season, making it easier to identify peak and off-peak rental seasons.

4. Scatter plot: This scatter plot compares rental prices with income levels in order to analyze affordability and income-to-rent ratios.

Rent vs. Property Size: This scatter plot illustrates the relationship between rental prices and property sizes, which aids in understanding how property size affects rentals.

5. Dashboard: A thorough analysis of rentals. Dashboard: A single dashboard that provides a brief overview of pricing distributions, rental trends, and key information. It has several charts and filters. Interactive exploration and analysis of the rental data is made possible by it.

By combining these methods and charts, real estate market players can gain a comprehensive understanding of rental data, allowing them to identify patterns and respond accordingly.

QUESTIONS:

1. In what ways do rental costs differ between US states or regions?

2. What are the long-term patterns in rental costs between 2010 and 2017?

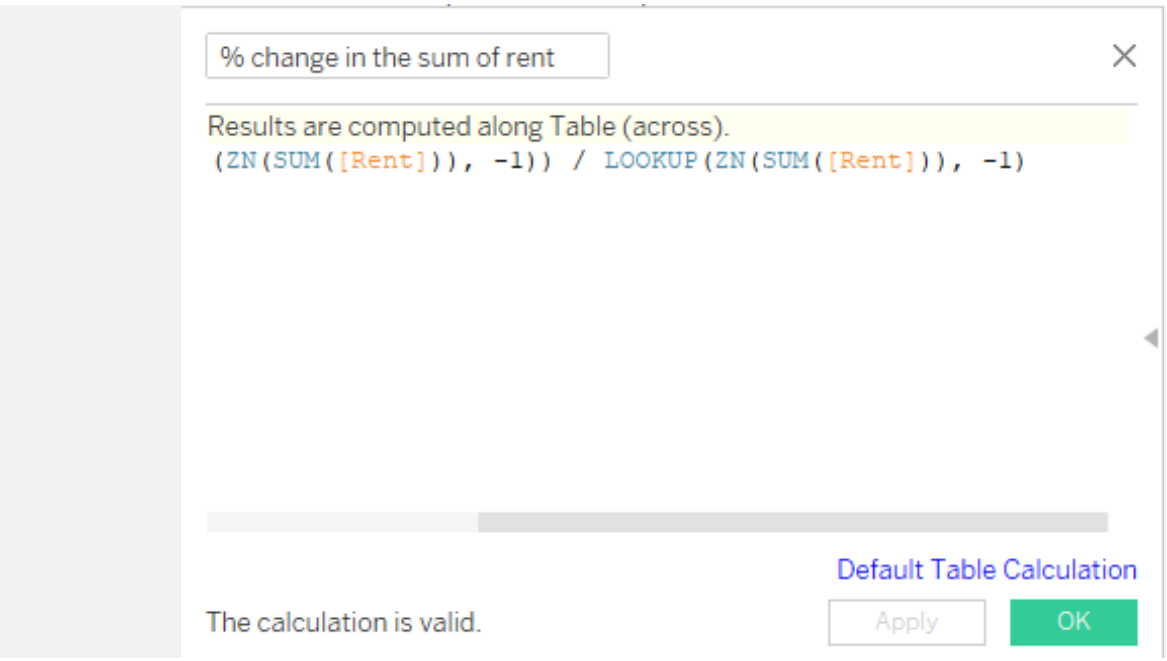
3. Do rental costs follow any seasonal trends or fluctuate?

4. How much does it cost per square foot based on population rank?

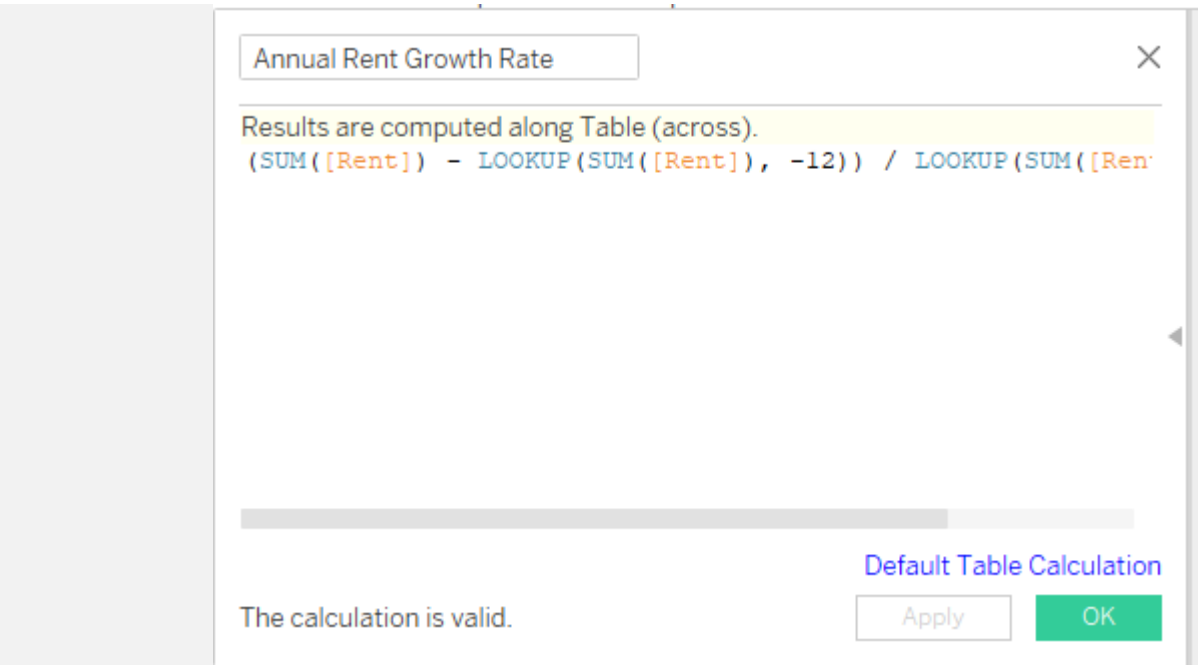
5. Michigan's average rental trend and projected rental prices for the future.

Calculated fields used:

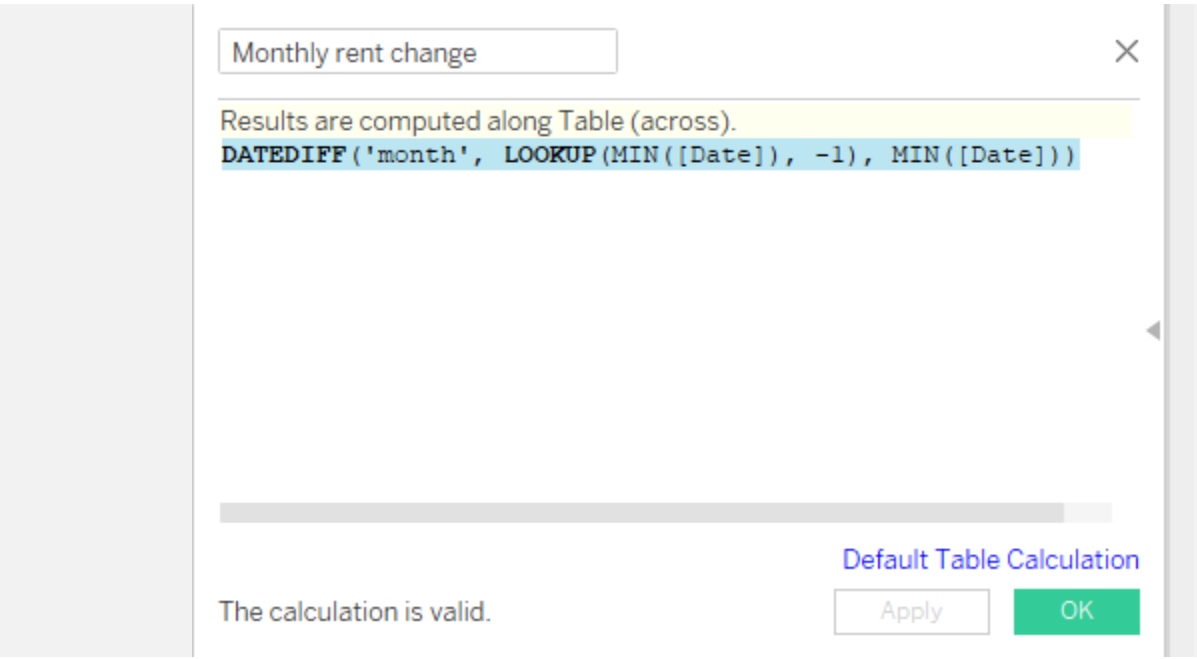
1. **% change in the sum of rent-** **(ZN(SUM([Rent])) - LOOKUP(ZN(SUM([Rent])), -1)) / LOOKUP(ZN(SUM([Rent])), -1)**



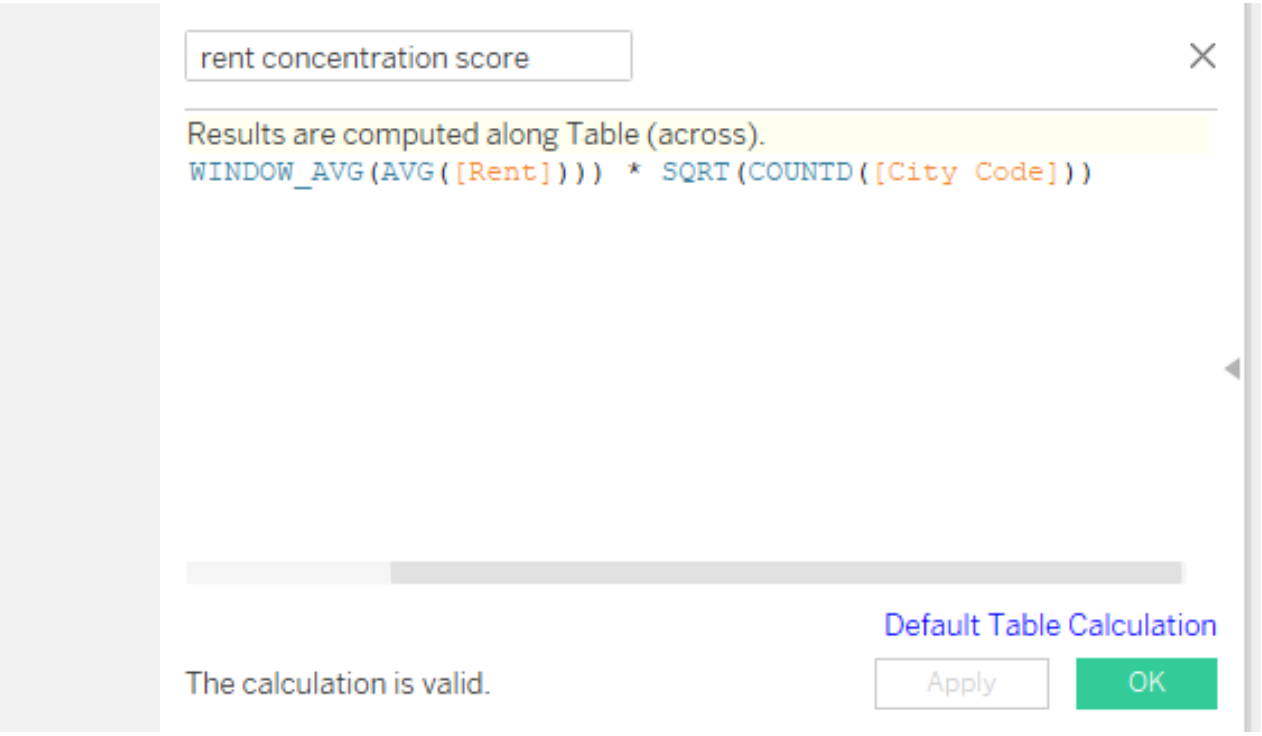
2. **Annual rent growth rate-** **(SUM([Rent]) - LOOKUP(SUM([Rent]), -12)) / LOOKUP(SUM([Rent]), -12)**



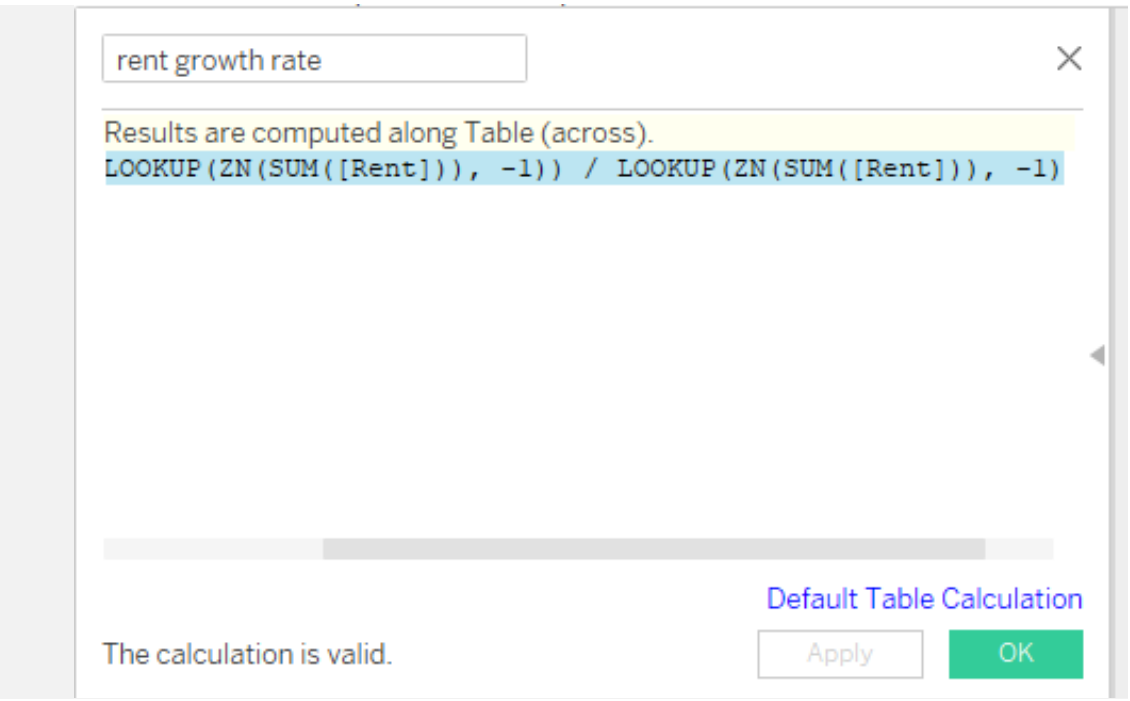
3.Monthly rent change- **DATEDIFF('month', LOOKUP(MIN([Date]), -1), MIN([Date]))**



4.Rent concentration score- **(AVG([Rent]) - WINDOW_AVG(AVG([Rent]))) SQRT(COUNTD([City Code]))**



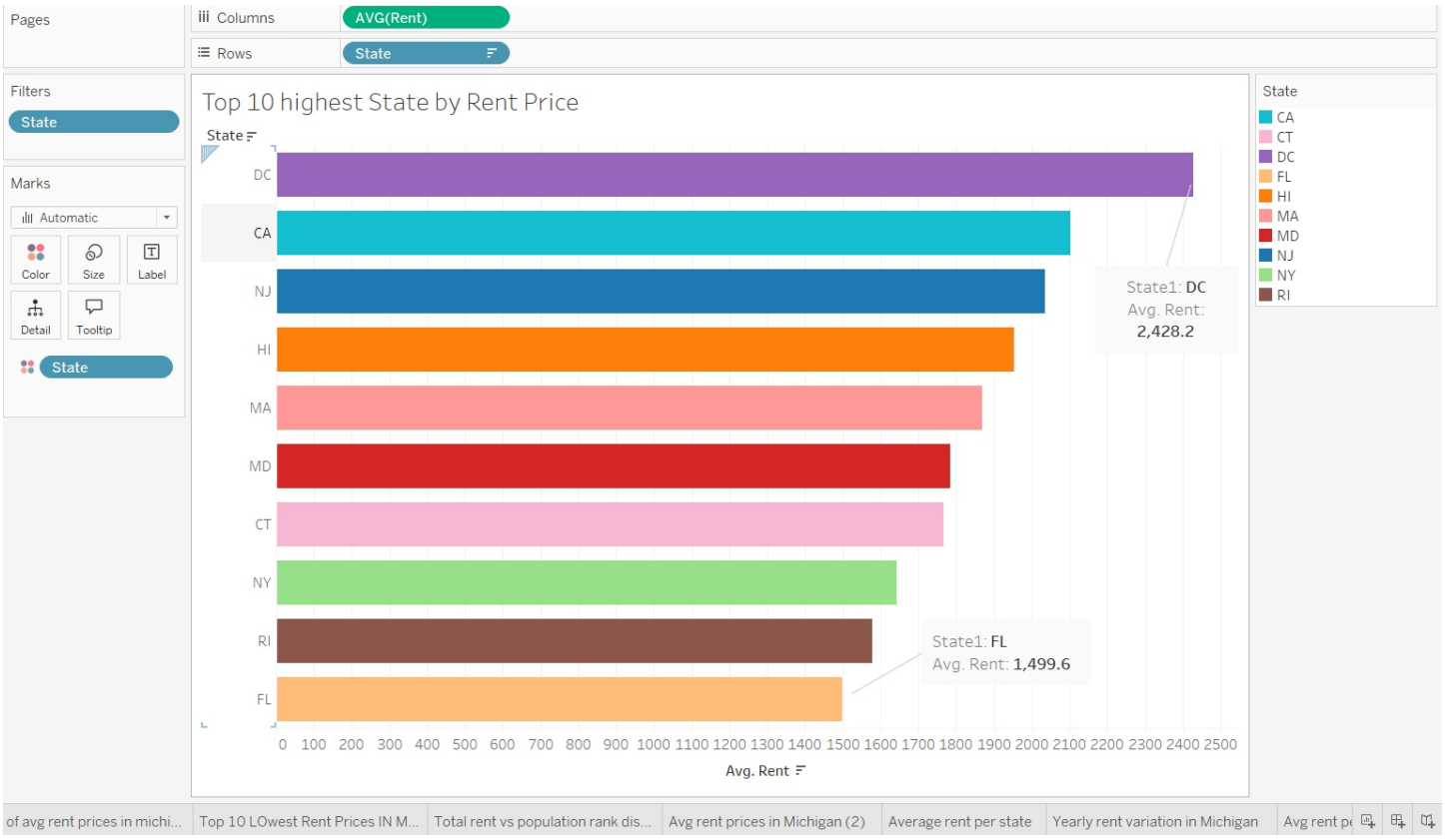
6. **Rent growth rate-** **(ZN(SUM([Rent])) - LOOKUP(ZN(SUM([Rent])), -1)) / LOOKUP(ZN(SUM([Rent])), -1)**



Visualization:

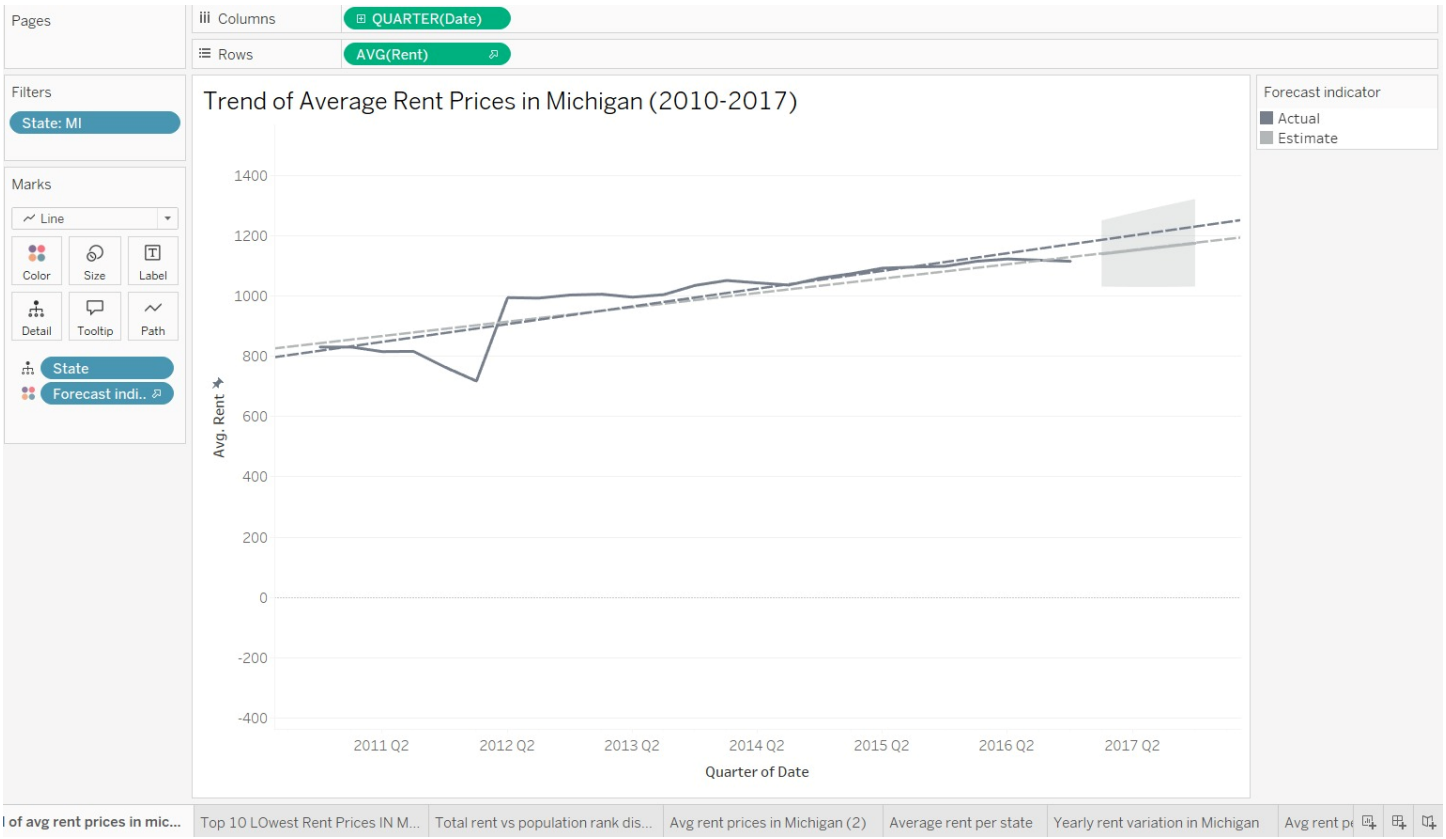
1. Average Rent by State:

- **Function used:**
  - AVG(Rent)
  - State on Rows
  - AVG(Rent) on Columns
  - Filter: Top 10 States by Rent
  - Color by State
- **Chart:** Horizontal Bar Chart
- **Result:** The top ten states in the US with the highest average monthly rent are displayed in this chart. Among the top states, Washington, DC has the most costly rent, at about $2,428 per month. With the lowest rent in the group—roughly $1,499 per month—Florida ranks tenth. The chart makes it simple to compare how expensive or affordable each state is for renters by using colored bars and obvious labels.



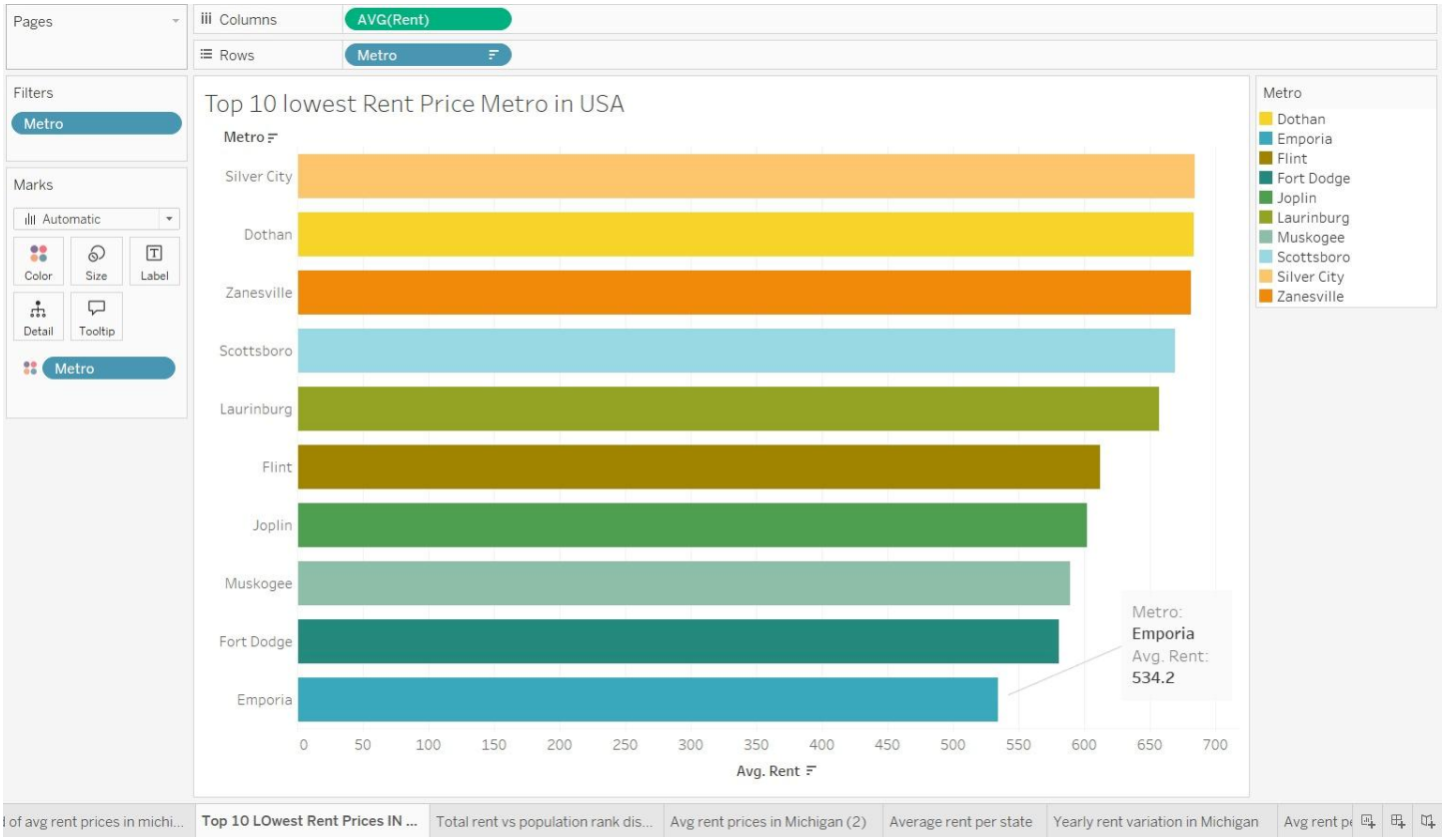
2. Analyze Trends – yearly rent variation in Michigan

- **Function used:**
  - AVG(Rent)
  - QUARTER(Date) on Columns
  - Forecasting enabled with Forecast indicator for Actual vs. Estimate
  - Filter applied: State = MI
- **Chart:** Line Chart with Forecast
- **Result:** The average rent in Michigan fluctuated every few months between 2010 and 2017, as depicted in this line graph. After a significant decline in early 2012, rent swiftly increased and remained stable in 2013 and 2014. It began to increase once more in 2015 and grew gradually until 2017. By the end of 2017, the average rent might surpass $1,200 per month, according to the prediction, which is based on historical trends.



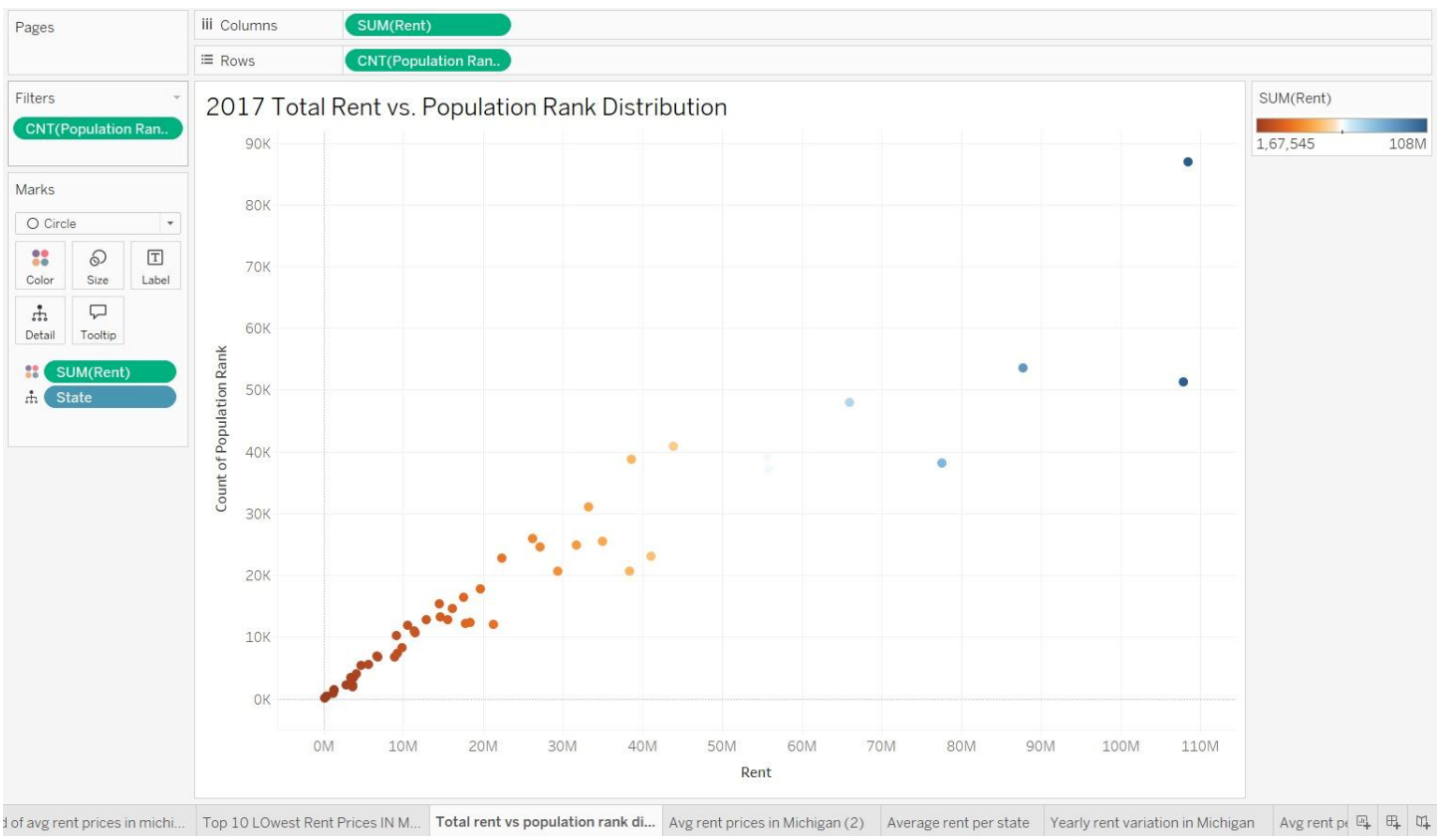
3. Top 10 Lowest Rent Prices in Metro

- **Function used:**
  - AVG(Rent)
  - Metro on Rows
  - AVG(Rent) on Columns
  - Filter applied: Top 10 metros with lowest average rent
  - Color by Metro
- **Chart:** Horizontal Bar Chart
- **Result:** The ten metropolitan areas in the United States with the lowest average rent prices are displayed in this chart. The lowest rent is found in Emporia, where it is approximately $534 per month. Fort Dodge, Muskogee, and Joplin also have relatively low rentals. When comparing the cost of living in various locations or searching for affordable places to live, this chart can be used to identify cities with reduced housing expenses.



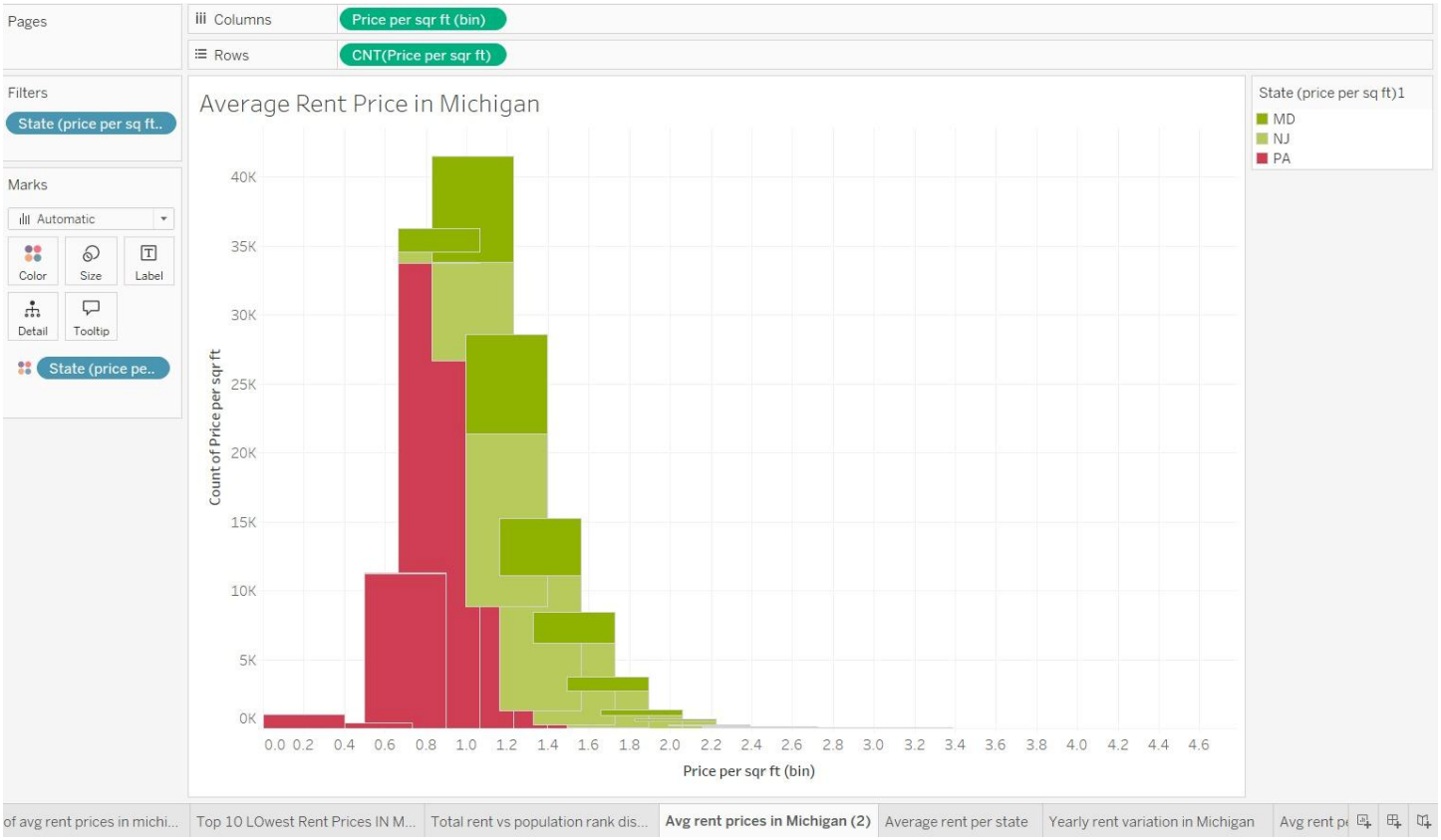
4. Total rent vs population rank distribution for the year 2017.

- **Function used:**
  - SUM(Rent) → Calculates total rent for each state or metro area
  - CNT(Population Rank) → Counts the number of occurrences for population rank per state
  - State used for color and detail
- **Chart:** Scatter Plot
- **Result:** According to this scatter plot, states with higher populations typically have higher overall rent values. That is, more persons typically translate into higher total rent payments. With a total rent of around $1.63 million and the highest population rank (1,260), Pennsylvania stands out the most. In addition to having large populations, New York and California both have high total rent. We can observe from this graphic how urban density and population size might affect each state's rent revenue.



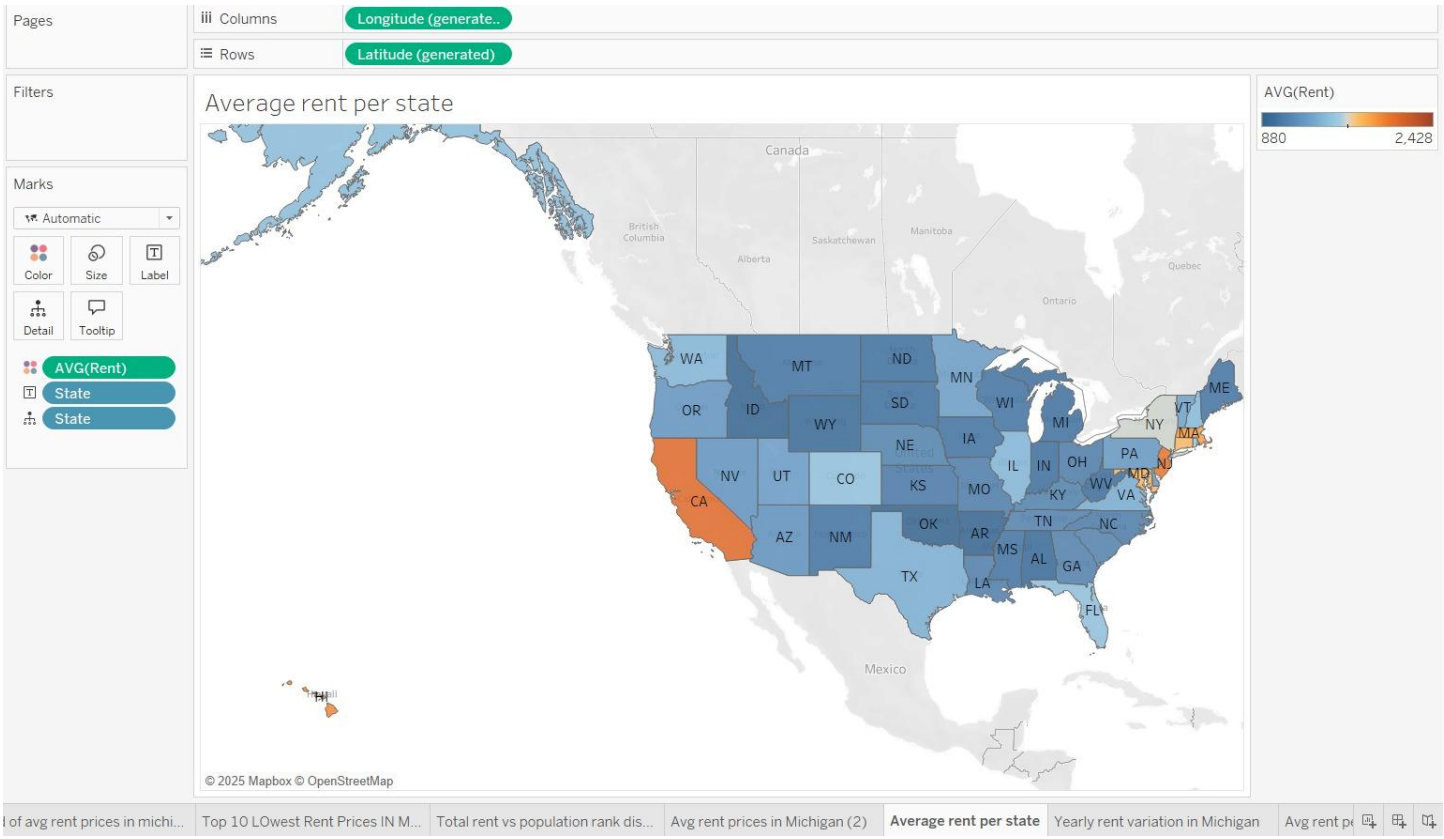
5. Average Rent Price in Michigan

- **Function used:**
  - Price per sq ft (binned for histogram)
  - CNT(Price per sq ft) → Count of occurrences per bin
  - State (used to compare MD, NJ, PA)
- **Chart:** Histogram
- **Result:** The distribution of rental costs per square foot in Maryland (MD), New Jersey (NJ), and Pennsylvania (PA) is depicted in this histogram. In all three states, the average rent is between $0.6 and $1.2 per square foot. Pennsylvania is often more economical because there are more rentals in the lower price bracket. Maryland and New Jersey, on the other hand, have more expensive options and a larger range of rent pricing, with some rentals ranging up to almost $2.2 per square foot.



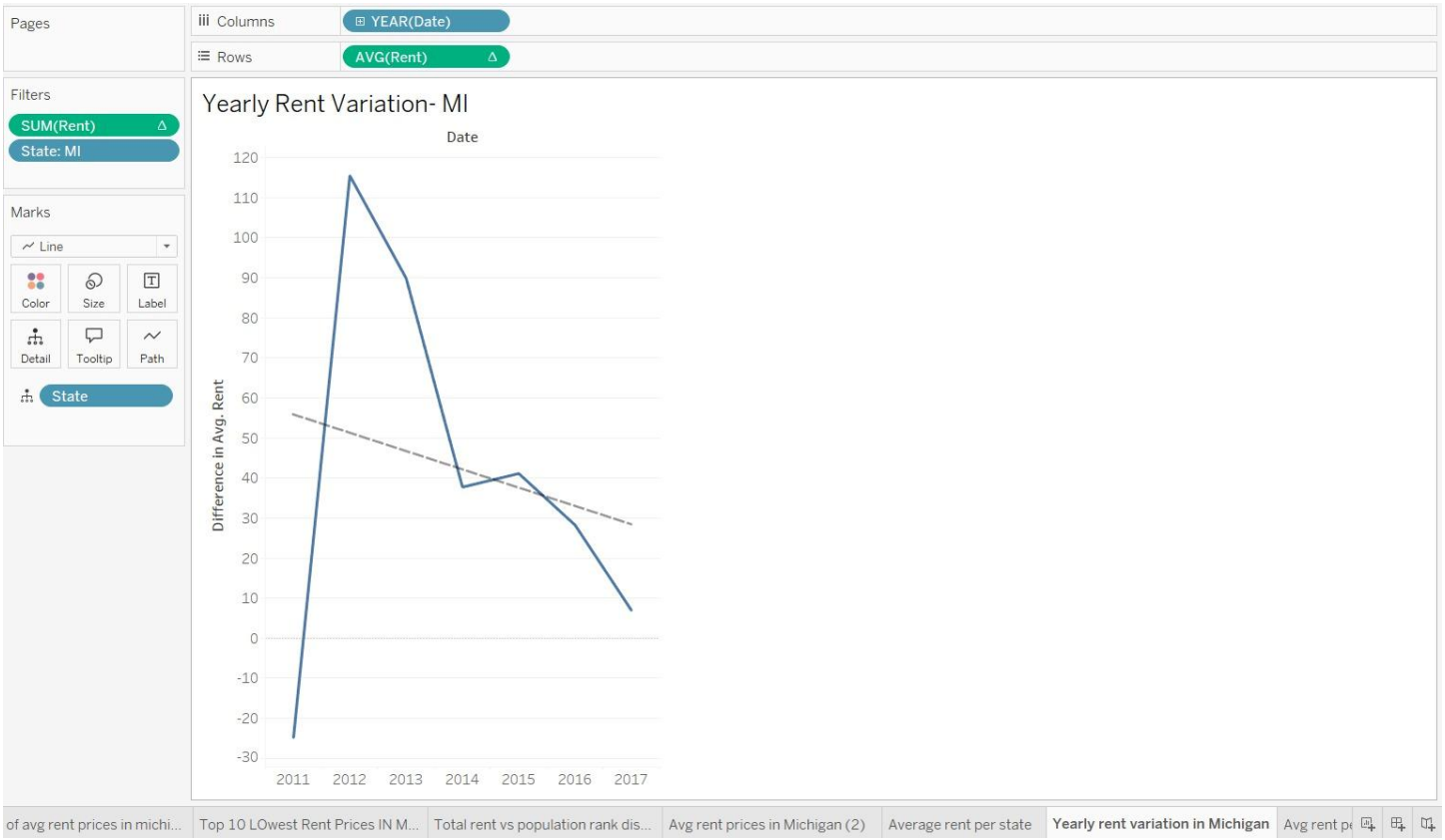
7. Average rent per state in the USA on Map

- **Function used:**
  - AVG(Rent)
  - State for geographic detail
  - Generated Latitude and Longitude for map positioning
- **Chart:** Filled Map (Choropleth Map)
- **Result:** Orange indicates states with the highest average rents (over $2,000), while blue indicates the majority of Midwestern and Southern states with lower average rentals (below $1,200). This suggests that home affordability in the US varies significantly by region.



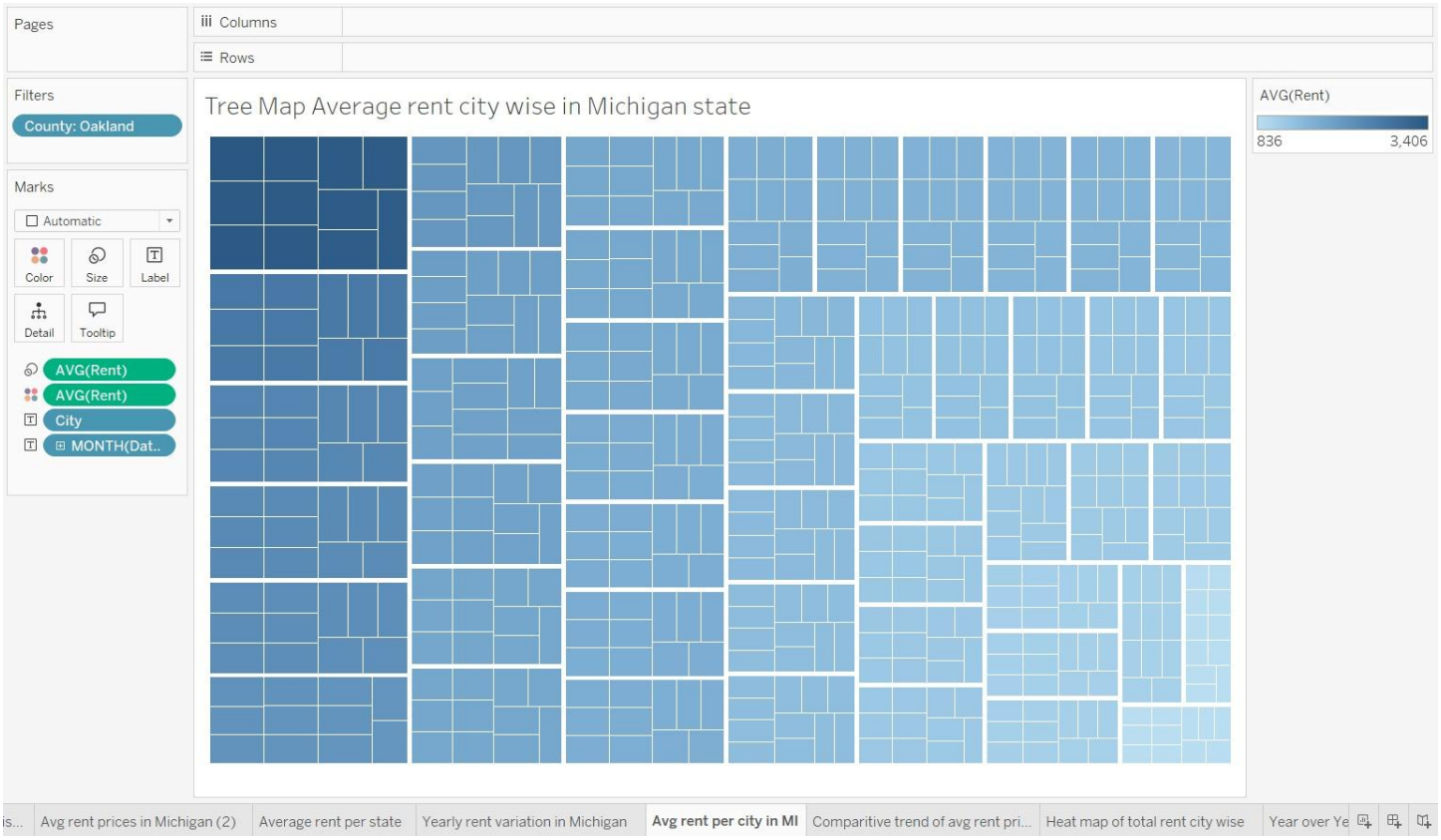
7. Yearly rent variation in Michigan

- **Function used:**
  - LOOKUP(SUM([Rent]), -1) → to calculate year-over-year rent difference
  - AVG(Rent) → displayed as line
  - YEAR(Date) → used for X-axis
  - Filter applied: State = MI
  - Trend Line → Linear trend added to show direction
- **Chart:** Line Graph with Trend Line
- **Result:** In Michigan, the average rent increased significantly in 2012 and then steadily decreased until 2017. If evaluated as the sum of year-over-year variations, the overall variance over the years seems to be 294.6, suggesting notable oscillations and a sustained downward trend in rent prices.



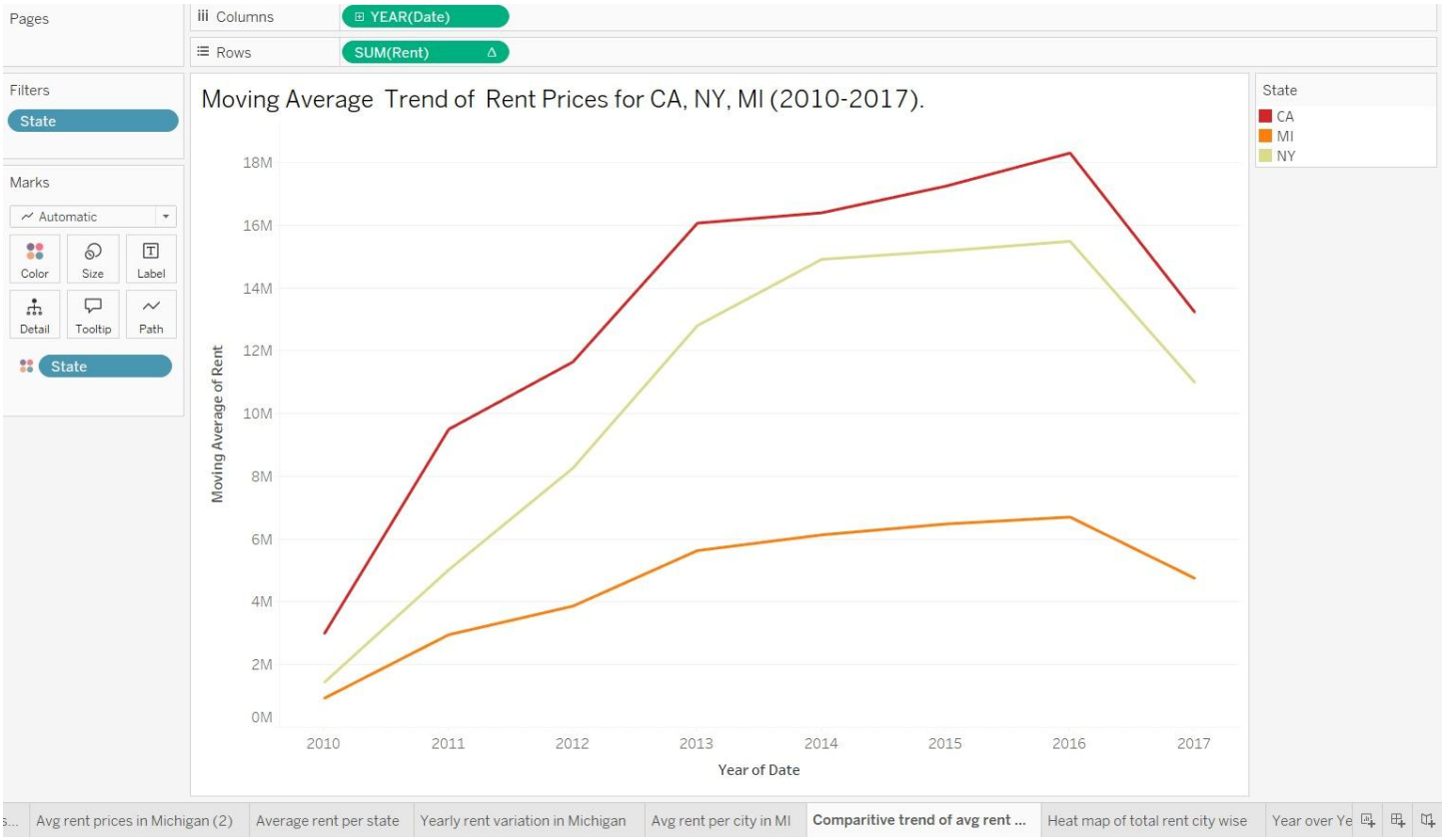
8. Average rent per city in Michigan State.

- **Function used:**
  - { FIXED [City], [State] : AVG([Rent]) }
  - AVG(Rent) dragged to Color and Size
  - Filter applied: County = Oakland
- **Chart:** Tree Map
- **Result:** The tree map shows Oakland County, Michigan's average rent by city. Larger blocks with deeper hues indicate cities with higher average rents. The average rent for all the cities in this view adds up to about $941, indicating that rental prices vary by city within the county.



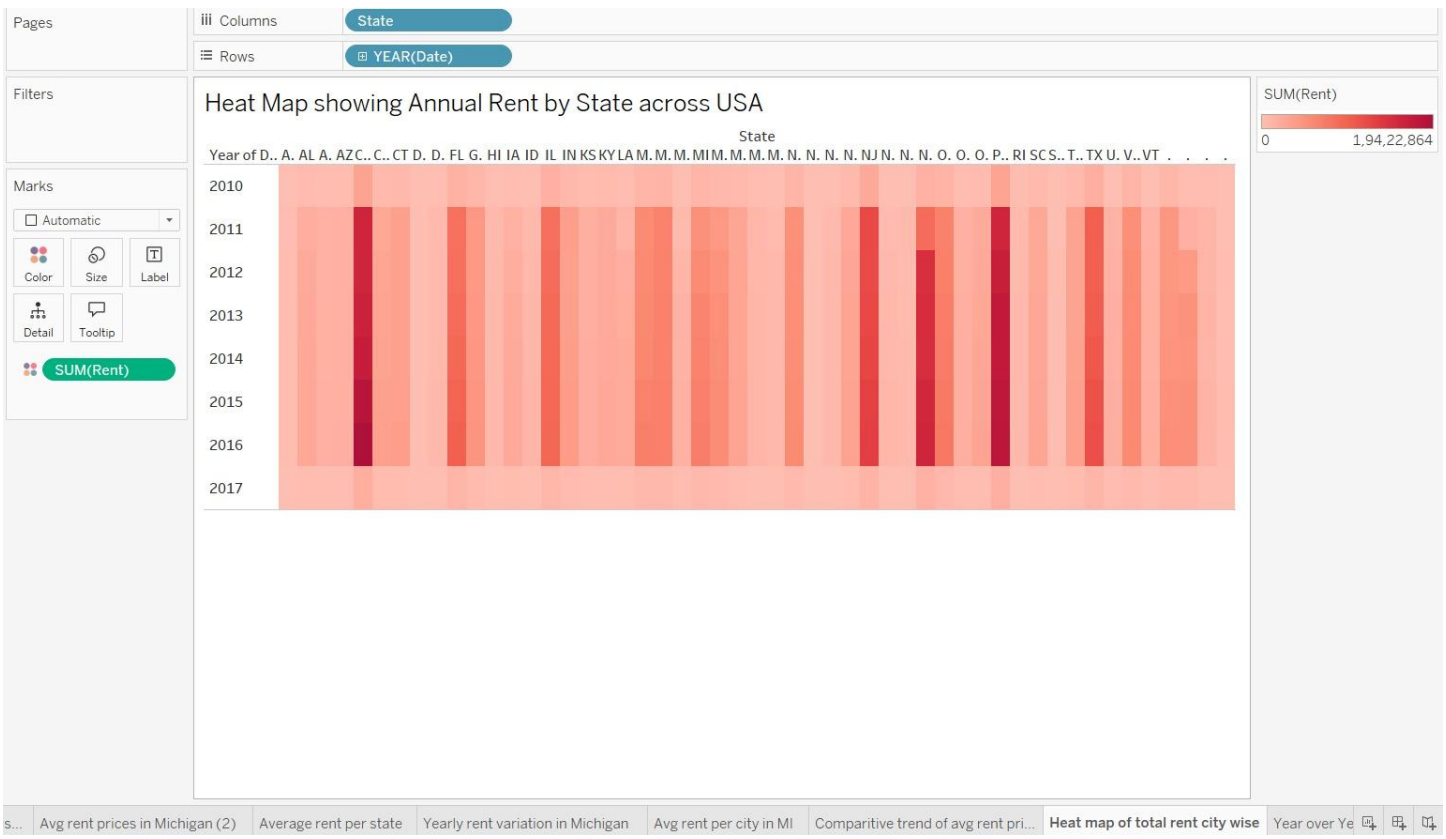
9. Comparative trend of avg rent prices in CA, MI, NY

- **Function used:**
  - SUM(Rent) over YEAR(Date)
  - Moving Average Table Calculation applied to SUM(Rent)
  - Filter: States = CA, NY, MI
  - Color by State
- **Chart:** Line Chart
- **Result:** California (CA) had the highest moving average rent from 2010 to 2017, with a peak of over $18 million in 2016–17. Next was New York (NY), which saw a sharp increase until 2015 and then a minor decrease. The lowest trend in rent was seen in Michigan (MI), which grew gradually before dropping after reaching a peak in 2016. This illustrates the notable differences in rent over time between the three states.



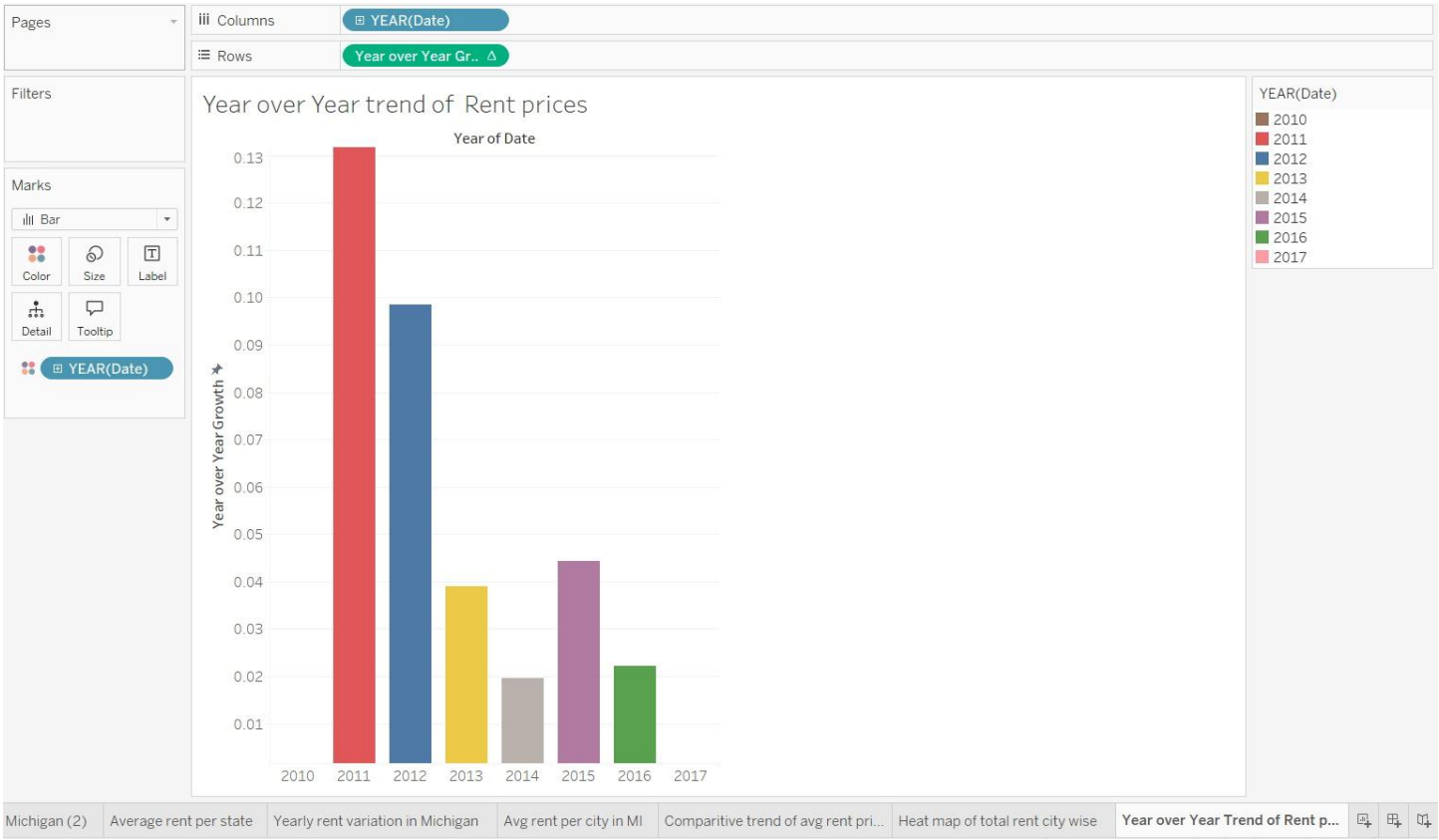
10. Heat map of total rent city wise across USA

- **Function used:**
  - SUM(Rent)
  - State in Columns
  - YEAR(Date) in Rows
  - Color encoding based on total rent per state-year combination
- **Chart:** Heat Map
- **Result:** The annual rent totals for each U.S. state from 2010 to 2017 are displayed on the heat map. States like California, New York, and Texas have the highest rent values, which are indicated by darker hues. The total amount of rent seen in this image, which represents cumulative rent activity across all states and years, is roughly $1,325,203.



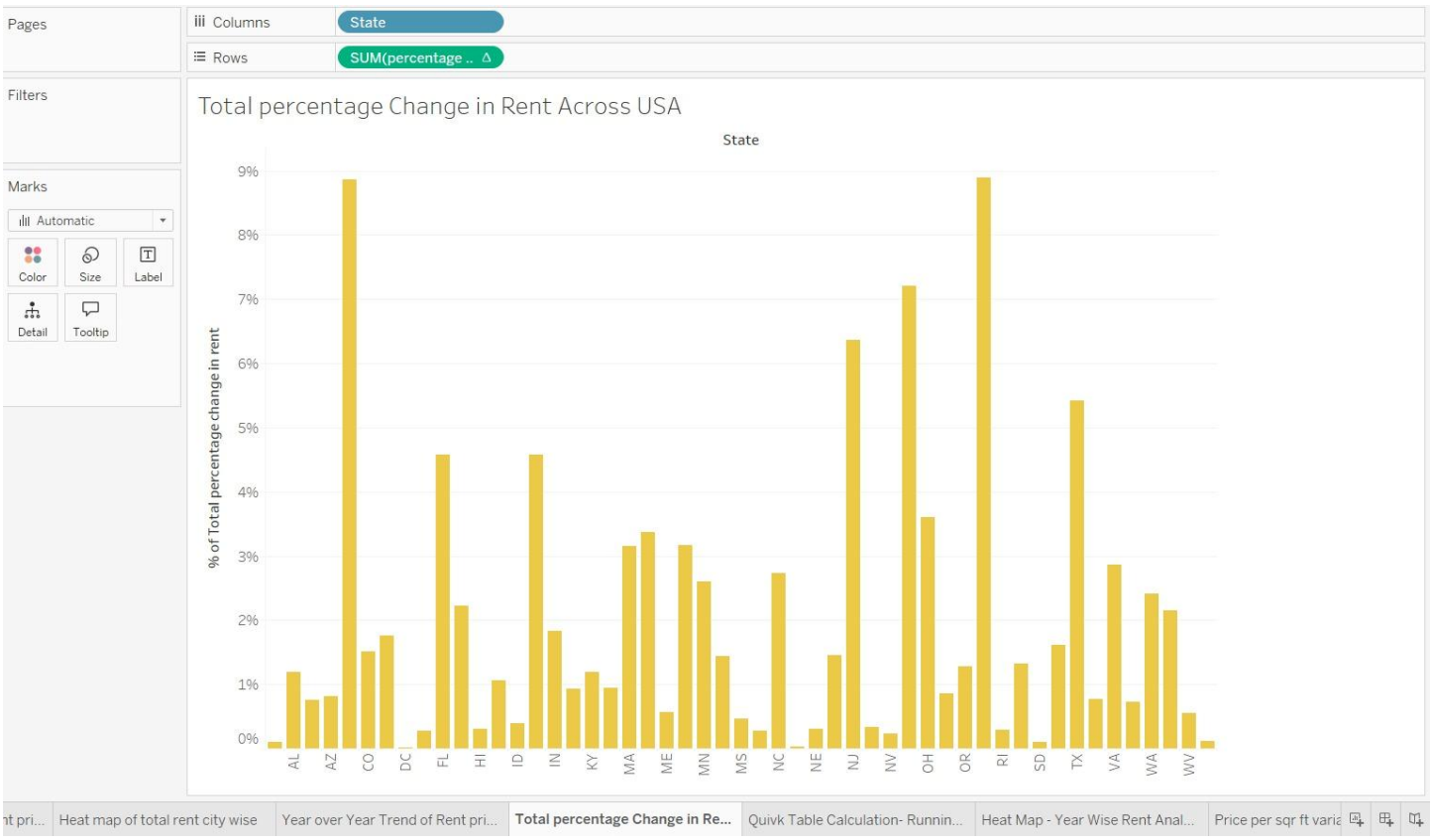
11. Year over Year Trend of Rent prices from 2010- 2017

- **Function used:**
  - YEAR(Date)
  - Year over Year Growth (Table Calculation on Rent)
  - SUM(Agg) → aggregated growth values
  - Color encoded by year
- **Chart:** Bar Chart
- **Result:** The graph displays the annual increase in rent prices between 2010 and 2017. 2011 saw the largest growth (13%), followed by 2012 (10%). Following that, growth rates gradually declined, suggesting a declining trend in rent rise. The overall cumulative increase throughout the period is represented by the sum of the YoY growth over the years, which is 4.465.



12. Total percentage Change in Rent Across USA

- **Function used:**
  - SUM (percentage)
  - State on Columns
  - Custom table calculation to compute % change across time (likely calculated using total rent over years)
- **Chart:** Bar Chart
- **Result:** The overall percentage change in rent by state is displayed in this chart. With an 8.895% increase, Pennsylvania (PA) had the largest gain, closely followed by California (CA) with an 8.853% increase. Colorado, New York, and Oregon are among the other states with rapid growth, which suggests that rents will rise significantly over the long run.

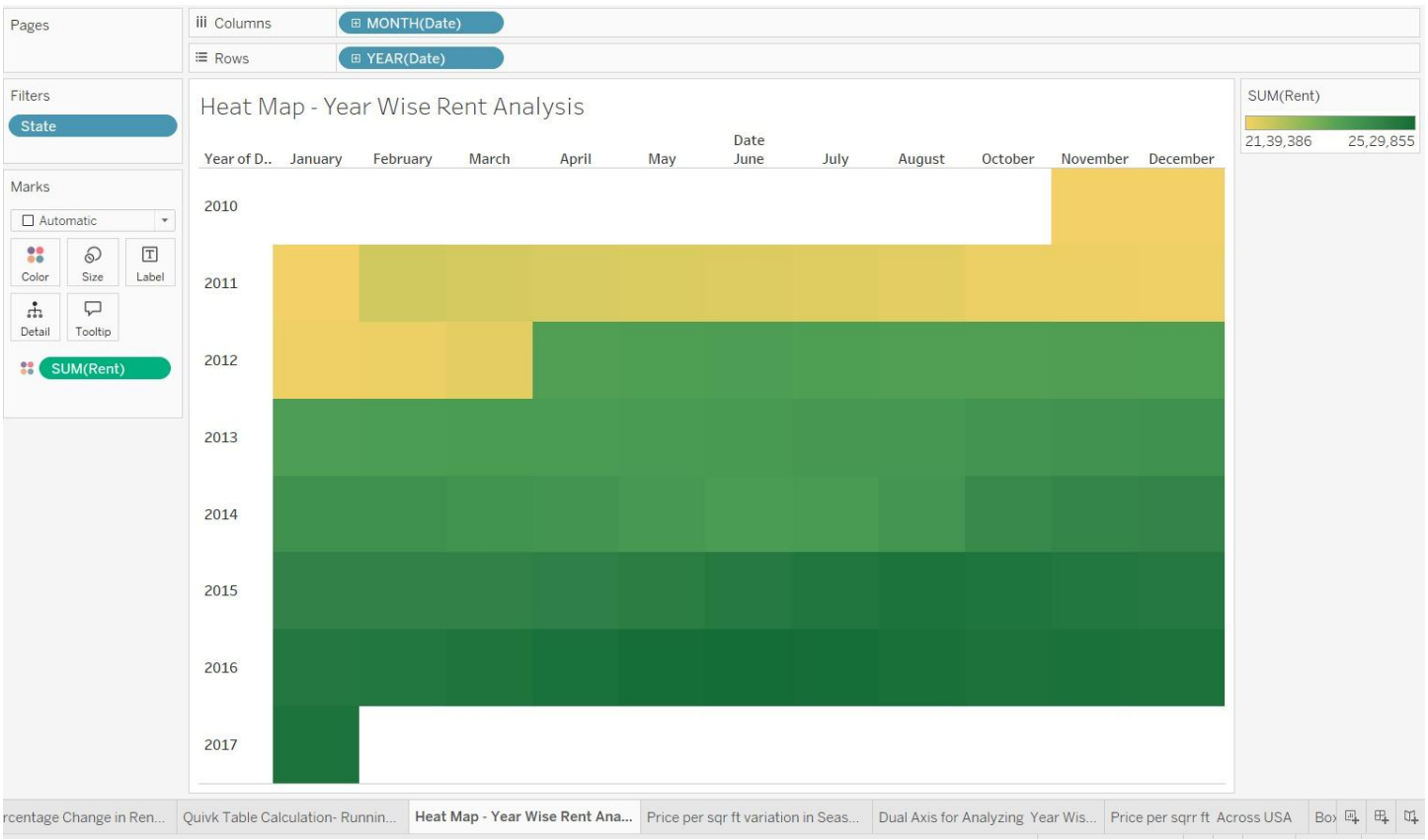


13. Quick Table Calculation- Running sum(Detroit Metro)

- **Function used:**
  - SUM(Rent)
  - Running Sum Table Calculation on SUM(Rent)
  - YEAR(Date) as X-axis
  - Filter applied: Metro = Detroit
- **Chart:** Dual Line Chart (1 line for yearly rent, 1 for running sum)
- **Result:** From 2010 to 2017, Detroit Metro's running sum of rent increased steadily, demonstrating cumulative rent growth over time. The annual rent, on the other hand, increased significantly between 2010 and 2011, was largely constant until 2016, and then started to fall in 2017—indicating a brief reduction despite long-term accumulation.

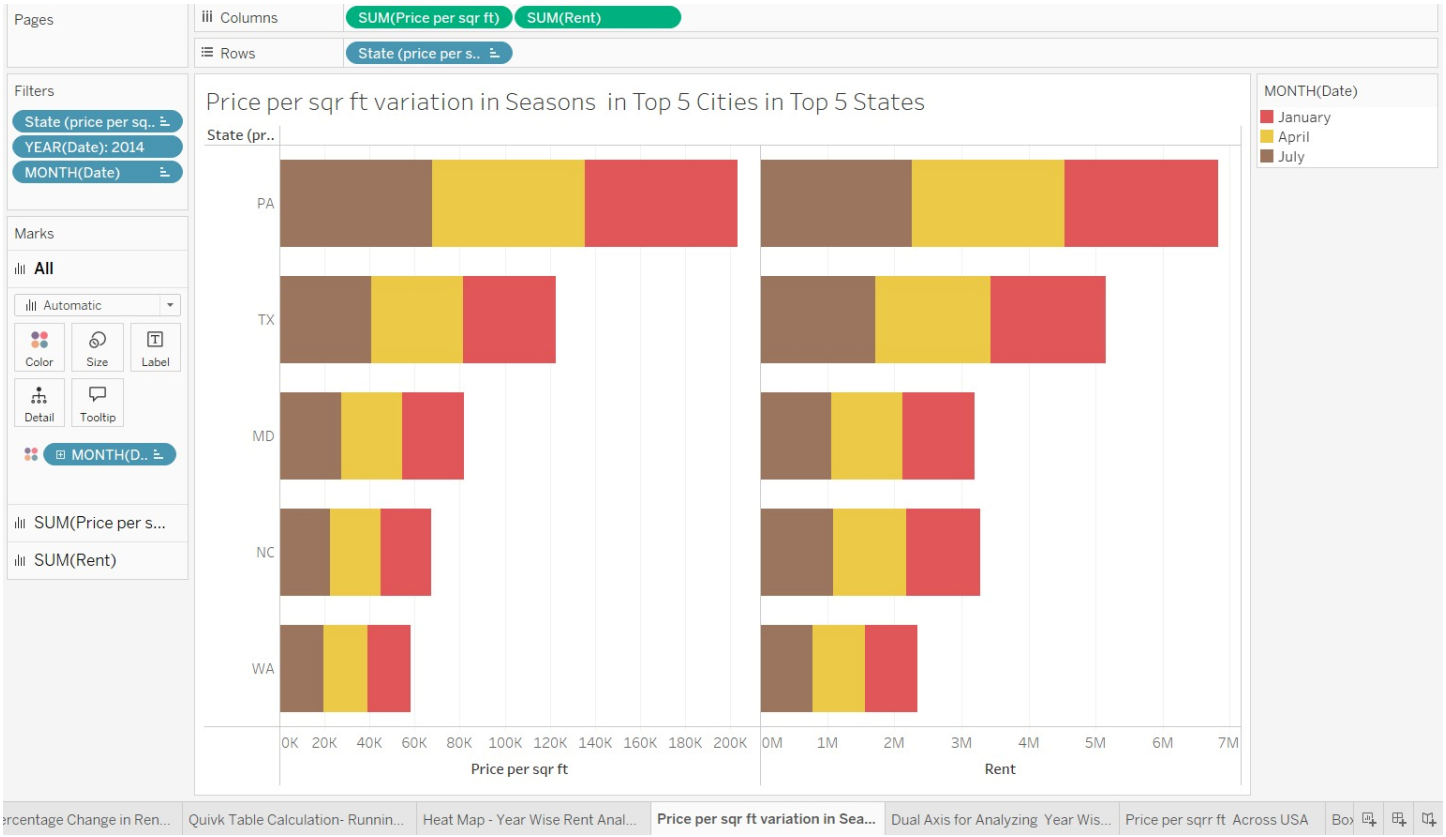
14. Heat Map - Year Wise Rent Analysis

- **Function used:**
  - SUM(Rent)
  - YEAR(Date) on Rows
  - MONTH(Date) on Columns
  - Color encoding by Rent amount
  - Filter applied: State
- **Chart:** Heat Map
- **Result:** This heat map displays the monthly changes in rent between 2010 and 2017. In the early years (2010–2012), the lighter hues had a lower mean rent. The colors become darker between 2013 and 2017, indicating a consistent annual increase in rent. The $178 million total rent for this time period demonstrates a consistent increase in rental costs over time.



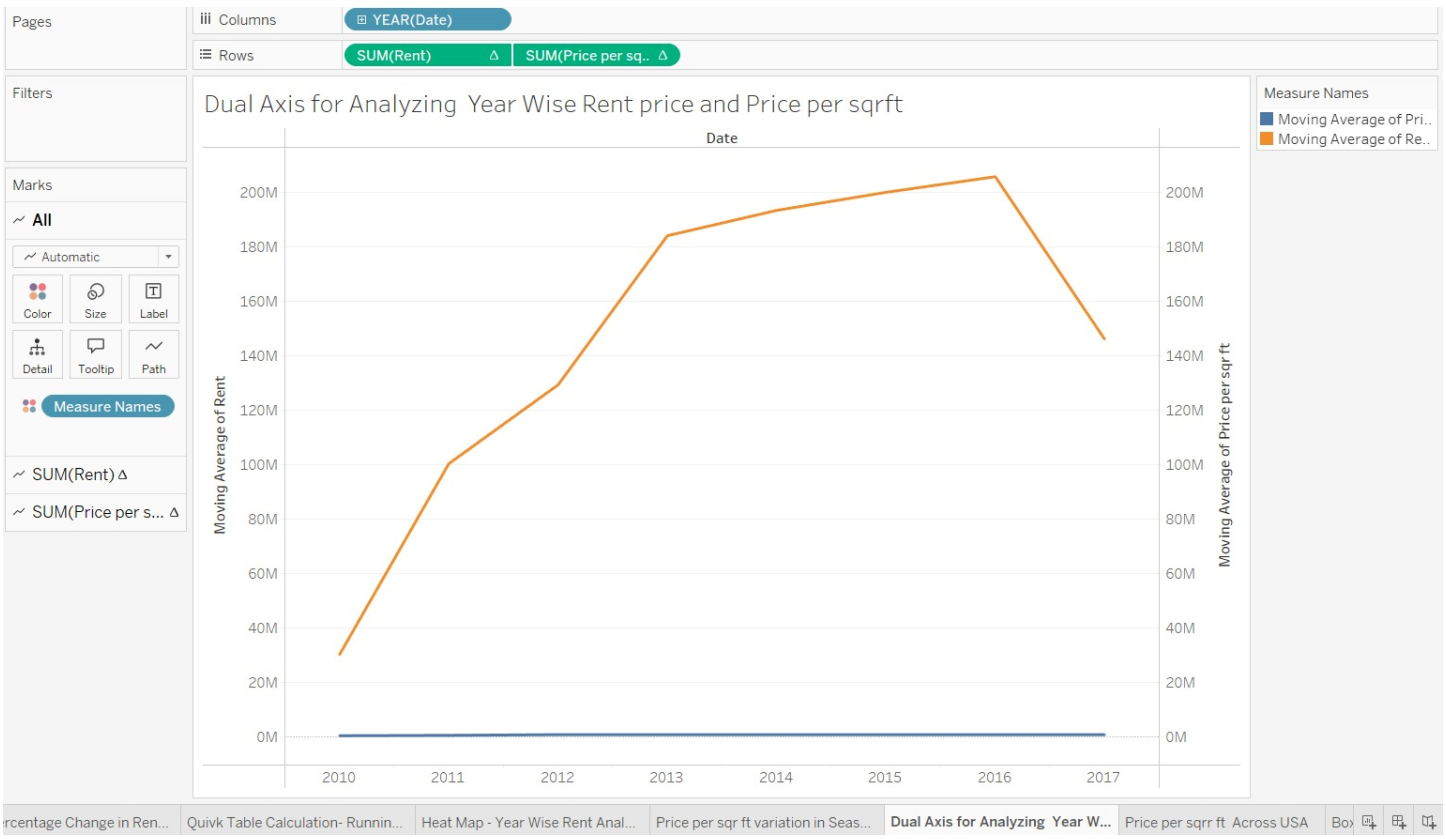
15. Price per sq. ft. variation in Seasons in Top 5 Cities in Top 5 States

- **Function used:**
  - State (price per sq ft)
  - SUM(Price per sq ft)
  - SUM(Rent)
  - MONTH(Date) for seasonal color grouping
  - Filter applied: YEAR(Date) = 2014
- **Chart:** Multiple Horizontal Bar Charts
- **Result:** The top 5 states—Pennsylvania, Texas, Maryland, North Carolina, and Washington—are shown in this graphic along with the seasonal variations in rent costs in January, April, and July. In each of the three months, Pennsylvania had the highest rent and price per square foot. There are noticeable seasonal variations in the price per square foot, which comes to 533.244 for all five states combined. Rent was typically higher in January and April than in July.



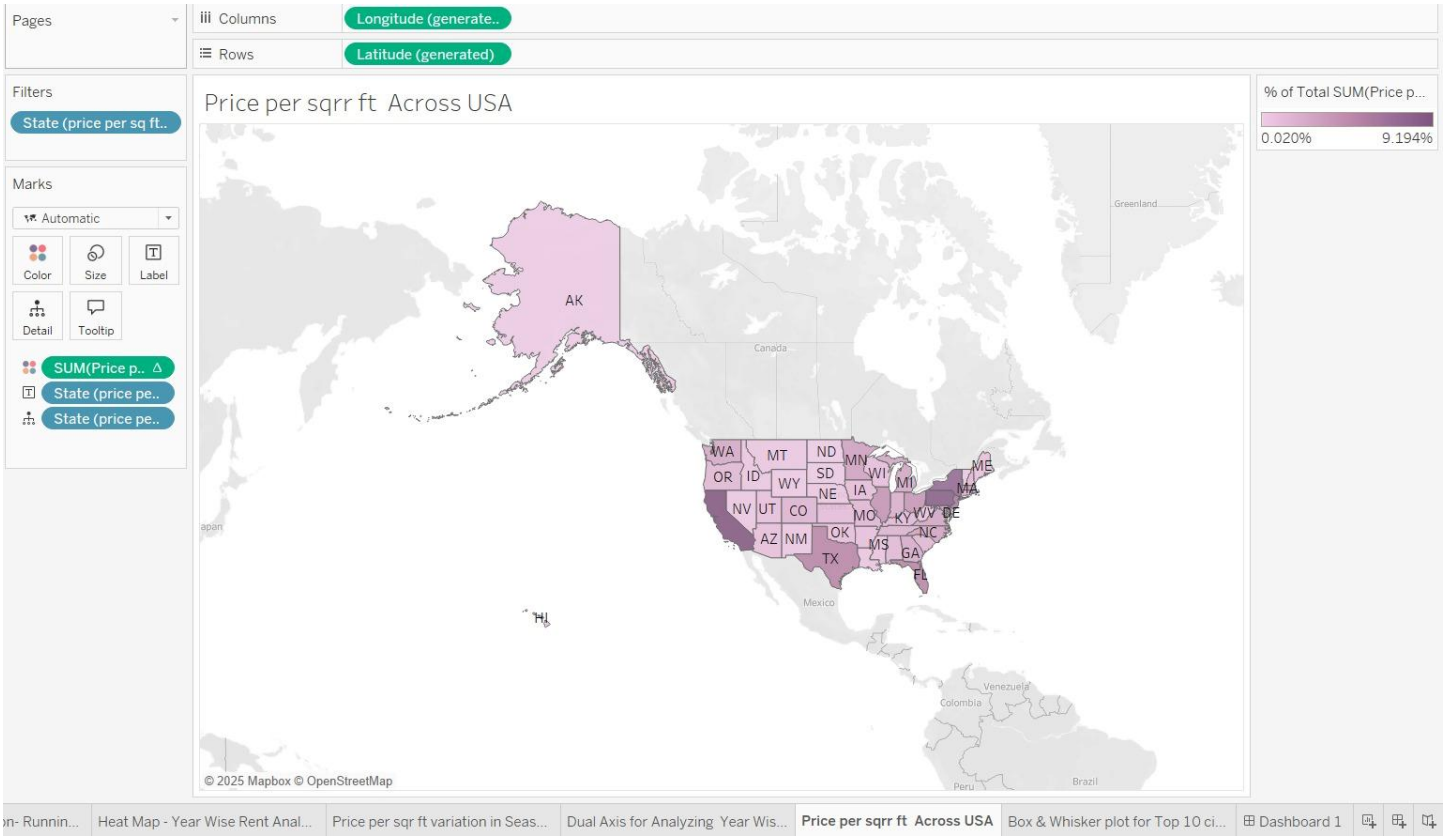
16. Dual Axis for Analyzing Year Wise Rent price and Price per sq ft

- **Function used:**
  - SUM(Rent)
  - SUM(Price per sq ft)
  - Moving Average Table Calculation applied to both measures
  - YEAR(Date) on X-axis
  - Measure Names used to create a dual axis view
- **Chart:** Dual Axis Line Chart
- **Result:** The moving average trends for both total rent and price per square foot from 2010 to 2017 are displayed in this chart. Total rent is represented by the orange line, which increases rapidly until 2016 before slightly declining in 2017. The blue price per square foot line remains relatively constant throughout time, indicating that costs per area haven't changed substantially. During this time, the average total price per square foot was roughly $5.6 million, indicating that although overall rent amounts fluctuated, the cost per square foot remained constant.



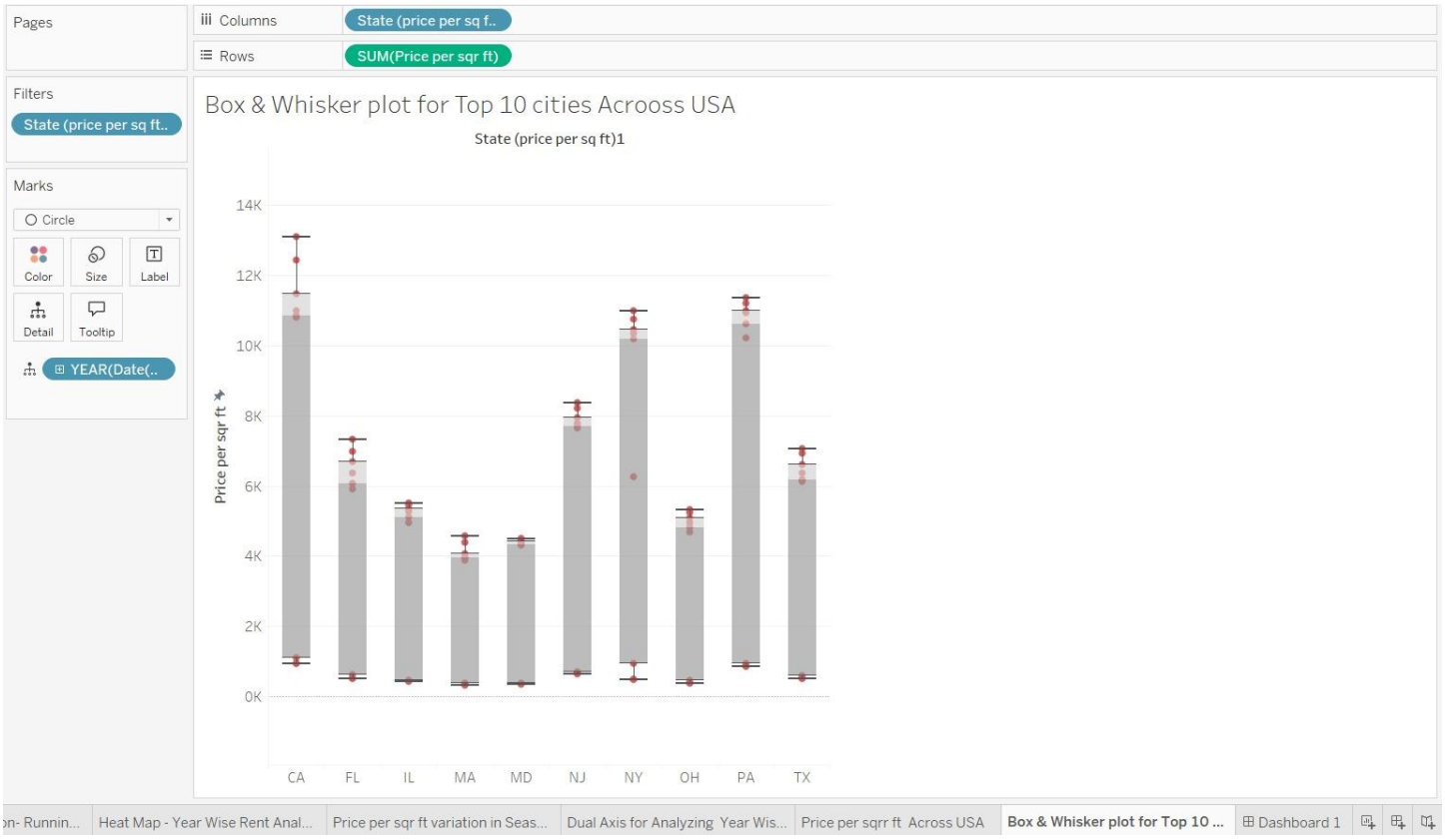
17. % of Price per sq ft Across USA

- **Function used:**
  - ([Price per Sq Ft] / WINDOW_AVG([Price per Sq Ft])) 100
    Calculates the percentage contribution of each state relative to the average price per sq ft across all states
  - Longitude and Latitude (Generated) for map placement
  - State in Detail and Label
  - SUM (Price per Sq Ft) on Color
- **Chart:** Map (Filled Map)
- **Result:** The amount that each U.S. state contributes to the national average rent price per square foot is displayed on this map. With a 9.2% share, California has the most impact, meaning that rental costs there are significantly higher than in other states. Lighter-colored states, such as South Dakota, Montana, and Alaska, on the other hand, have a negligible effect (around 0.02%), indicating that rent costs are far lower there.

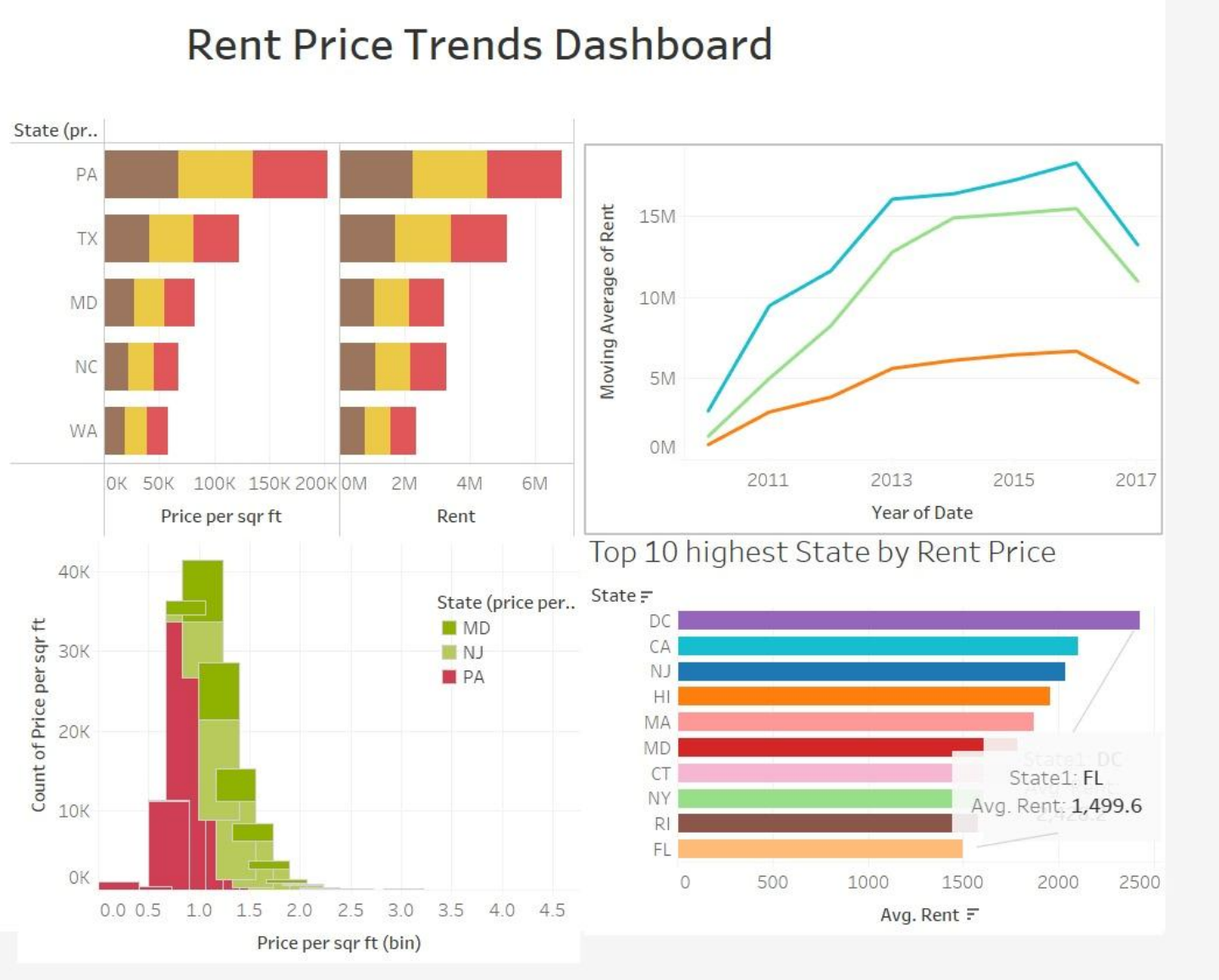


18. Box & Whisker plot for Top 10 cities Across USA

- **Function used:**
  - [Price] / [SquareFeet] → to compute Price per Sq Ft
  - SUM(Price per sq ft) used on Y-axis
  - State (price per sq ft) on X-axis
  - Filter: Top 10 cities by price per sq ft
  - Breakdown by YEAR(Date) to plot individual points
- **Chart:** Box & Whisker Plot
- **Result:** This graph displays the distribution of rent prices per square foot in ten states' main cities. There is a great deal of variability and some extremely pricey places in California and New York, which have the largest range and the highest prices. A significant amount of expensive real estate expenses is concentrated in major cities, as evidenced by the sum of the prices per square foot for all of these cities, which comes to 450,357.



19. DASHBOARD



Conclusion and Suggestions:

Trends Found: The data revealed both short-term swings and long-term trends in rental pricing. Businesses in the property management and real estate industries need to be aware of these trends.
The top five cities for annual rent increases are:
Among all types of rental housing:

- Providence, R.I.: +8.2%
- Louisville, Ky.: +6.9%
- Cleveland, Ohio: +6.5%
- Hartford, Conn.: +6.4%
- Boston: +6.1%

Of all rental property types, the cities with the top five yearly rent price decreases or the smallest rises are:

- Austin, Texas: -3.3%
- San Antonio, Texas: -0.5%
- Portland, Ore.: +0.1%
- Dallas: +0.2%
- Atlanta: +0.5%

Seasonal Patterns: From 2010 to 2017, rental costs in the US varied according to the seasons. Since many people travel and rent homes for holidays, the demand for rentals was typically higher in the summer, particularly in well-known tourist destinations. On the other hand, demand was frequently reduced during the winter, particularly in colder climates where fewer people travel or relocate. However, these patterns may differ dependent on local conditions and the time of year, so it's critical to modify rental prices accordingly.

Regional Disparities: During this time, rental rates also differed from city to city in the United States. Both property owners and rental companies might benefit from these distinctions when developing their marketing plans. For instance, they could concentrate on marketing homes in high-demand locations or change prices in areas with lower rental costs. Businesses may target the proper audience and set prices that reflect the local market by knowing how rent costs vary by location.